Course Number: PSP4706	***Course Name:*** Logical Reasoning & Quantitative Analysis	BML MUNJAL UNIVERSITY FROM HERE TO THE WORLD
Credits: 1 (0-2-0)	***Contact hours per week*** = 2 hours	
Course Faculty: Mr. Nitesh Kumar **Course Coordinator:** Dr. Kiran Khatter	**Email:** v-niteshkumar.hbe@bmu.edu.in kiran.khatter@bmu.edu.in	

Aim of the course: The primary goal of this course is to equip students with the essential skills required for logical reasoning and quantitative analysis, preparing them for placement tests conducted by reputable companies. The course aims to develop proficiency in logical reasoning and quantitative problem-solving techniques, which are crucial for success in the screening process for various job positions in the corporate world. Through comprehensive instruction and practice, students will gain the ability to effectively tackle technical and logical challenges commonly encountered in placement tests and interviews.

Course Overview and Context: - The central focus of this course is to impart a profound comprehension of logical reasoning and quantitative analysis, emphasizing their practical applications in preparation for placement tests conducted by renowned companies. Through a structured curriculum, students will delve into the intricacies of logical reasoning techniques, quantitative problem-solving methodologies, and algorithmic approaches. This course seeks to empower students with the essential skills required to excel in placement tests, interviews, and real-world scenarios by providing a solid foundation in the core principles of logical reasoning and quantitative analysis.

Topics of the Course

Numbers & LCM: This topic covers the properties of numbers and focuses on the calculation of the Least Common Multiple (LCM) of two or more numbers.

Average & Decimals: This section deals with calculating averages and working with decimal numbers, which are essential in various quantitative calculations.

Algebra: Algebra involves solving equations, manipulating variables, and studying algebraic expressions, providing a foundation for more advanced mathematical concepts.

Percentage: This topic explores the concept of percentages, teaching how to calculate percentages, find percentage change, and apply percentages in various real-life situations.

Profit & Loss: Profit and loss concepts help individuals understand how to calculate profits or losses in business transactions and make informed financial decisions.

Mensuration & Geometry: Mensuration involves the measurement of geometric shapes and figures, while geometry explores the properties and relationships between shapes and angles.

Work & Chain Rule: Work and chain rule problems deal with rates of work and the interconnectedness of tasks.

Ratio & Partnership: This topic delves into ratios, proportions, and partnership agreements, useful for understanding the distribution and sharing in various scenarios.

Time, Speed & Distance: Time, speed, and distance problems involve calculating distances traveled at different speeds and the time taken to reach destinations.
Permutation & Combination, Probability: Permutation and combination concepts are about counting and arranging objects, while probability deals with the likelihood of events occurring in various situations.

Clocks & Calendars: This section covers topics related to timekeeping using clocks and calendars, including calculations involving time intervals and date-related questions.

Venn Diagram & Logs: Venn diagrams are used to visualize and analyze data relationships, while logarithms (logs) are mathematical functions with various applications in solving equations and modeling exponential growth.

Series – Numerical & Alphabet & Coding-Decoding: This topic involves identifying patterns and sequences in numerical, alphabetical, or coding-decoding series, helping in logical and analytical thinking.

Blood Relations & Directions: Blood relations explore familial relationships, while directions involve understanding and navigating based on cardinal directions (north, south, east, west), crucial for logical reasoning and spatial awareness.

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Analytical Reasoning - Selection: Analytical reasoning in the context of selection involves making choices based on given criteria, often seen in competitive exams and job assessments.

Syllogism: Syllogism is a type of logical reasoning that deals with deducing conclusions from given premises, enhancing deductive reasoning skills.

Course Outcomes:
At the end of the course, the students should be able to:

CO1: Understanding of fundamental principles of logical reasoning and quantitative analysis
CO2: Apply logical reasoning and quantitative analysis techniques to solve diverse and complex problems CO3: Analyze and break down complex problems to identify patterns, relationships, and solutions.

CO PO Mapping

Three Levels of Mapping: 1: means weak, 2: means medium, and 3: means strong.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	POS4
CO1					**2**	**3**							**1**		**2**	
CO2	**2**	**2**	**3**		**2**			**2**					**2**		**1**	
CO3		**2**			**2**								**2**		**1**	

Session Plan:

Course Topic	No of sessions
Numbers & LCM	1
Average & Decimals	1
Algebra	1
Percentage	1
Profit & Loss	1
Mensuration & Geometry	1
Work & Chain Rule	1
Ratio & Partnership	1
Time, Speed & Distance	1
Permutation & Combination, Probability	1
Clocks & Calendars	1
Venn Diagram & Logs	1
Series – Numerical & Alphabet & Coding-Decoding	1
Blood Relations & Directions	1
Analytical Reasoning - Selection	1
Syllogism	1

Learning Resources:

The students will have access to LMS which will have the below-stated modules available to the students:

- Aptitude Basic
- English Communication
- Aptitude Advanced
- Interview Preparation
- Engineering Capsule
- Company Specific Test Series
- Test Series and Result Analytics

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Experiential Learning and Assessment:

In this course, students will work on scenario-based questions and problems that reflect real-world challenges. They will develop practical applications by improving existing methods or adapting them to diverse contexts. Students are expected to demonstrate results through implementation and can also explore integration with concepts from other courses, such as Digital Image Processing or Artificial Intelligence. A separate evaluation will assess their proposed solutions, focusing on:

- Critical thinking to address complex and uncertain scenarios.
- Application of techniques to solve real-world problems.
- Skills in articulating technical concepts effectively.

The final grade will be based on the marks/ grades obtained in the mid-semester and end-semester evaluation and other assessments defined in the assessment table. The relative grading method described in the university's academic regulations will be followed to grade the students. The student must secure a minimum of 40% of marks after completing all the assessments in the following table to become eligible for grading.

Component	**Weightage (%)**	**Evaluation Week**	**Remarks**
Class Participation	10%	Continuous	- Based on assignments given and level of participation in the discussion
Evaluation -1	25%	In the week starting from 17 March 2026/as per the mid-term date sheet	Students will apply logical reasoning and quantitative analysis techniques to solve complex MCQs and practice questions on an online portal (conducted by faculty), with evaluation conducted to enhance problem-solving and critical thinking skills.
Evaluation -2	25%	Last week of April 2026	
End Term Evaluation	40%	In the last week of the semester/ as per the end-term date sheet	

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Prepare for discussions on the analysis part of Lab Assignments during class every week.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

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Recourse examination policy: As per University policy.
Make-up policy: No make-up work will be given for unexcused absences. The faculty needs to be informed in advance if the student is going to miss submitting an assignment or taking any evolution component. It is at the discretion of the faculty to sanction make-up for an evaluation component.
Behavior expectations: No mobile phones and other distractive gadgets are permitted in the class. Laptops are allowed during the defined course for assignment only.
Academic dishonesty/cheating/plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.

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